The finite terms of `Group_I/II/III.tex`: everything that is not $\propto \log\frac{k^+}{\Lambda}$ or $\log\frac{\vee}{k^+}$, with $\Lambda \to 0$

Companion of `LL_terms.pdf`. Each row is $\log\frac{k^+}{\Lambda}\,\mathfrak{T}_{\rm row} + \log\frac{\vee}{k^+}\,\mathfrak{P}_{\rm row} + R_{\rm row}$ exactly; here is $R_{\rm row}$, row by row, in your form. Blue: new; red: removed or cancelling.

1 Rules and summary

- Each logarithm is replaced by its remainder (`LL_terms.pdf`, Sec. 2): $\left[\log\frac{k^+}{\Lambda} + \log(\frac{\vee}{k^+} - 1)\right] \to \log(1 - \frac{k^+}{\vee})$, $\log\frac{\vee}{\Lambda + k^+} \to 0$, $\log\frac{(\vee - k^+)A + k^+B}{\Lambda A + k^+B} \to \log\frac{(1-\frac{k^+}{\vee})A + \frac{k^+}{\vee}B}{B}$, etc.
- $\Lambda \to 0$ wherever possible (the exact remainders differ by terms that vanish for $\Lambda \to 0$: 3.5×10^{-8} at $\Lambda = 10^{-9}$); $\vee$ is kept exact.
  - In the limits of the plus-prescribed integrals of Group I Row V 2b and Group II Row IV: $\int_\Lambda^{k^+-\Lambda} \to \int_0^{k^+}$, $\int_{k^+-\Lambda}^{\Lambda} \to \int_{k^+}^0$, $\frac{k^+}{k^+ + \Lambda} \to 1$.
  - $\mathcal{A} \to \mathcal{A}_0$,

$$\mathcal{A}_0 = \frac{1}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \left\{ \arctan\left[\frac{(x-z)^2 - (x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right] + \arctan\left[\frac{(x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right]\right\} \tag{1.1}$$

- **Hidden logarithms** (Group II Row IV, Group III Rows III, V, VI): $\left[\log\frac{k^+}{\Lambda} + \log(\frac{\vee}{k^+} - 1)\right] \to \log(\frac{\vee}{k^+} - 1)$, which cancels the $\log\frac{\vee}{k^+}$ of the $\left[\frac{1}{p^+}\right]_+$ (resp. $\frac{\xi}{(1-\xi)_+}$) integral; $\left[\frac{1}{p^+}\right]_+$ is kept unevaluated. The BFKL term contained in that integral is subtracted (blue, C1, C3); where both come from: Sec. 2.
- **Cancelling lines** (red): the $\frac{1}{\Lambda}$ lines of Group I Row III (sum zero, 3.0×10^{-16}), your two red lines of d^3N_2, and the edge-kernel line of d^3N_{3b} (C4, Q3).
- **C5**: the measure of Group III Row III part 2.
- **Running coupling and DGLAP** are in Group I Row II part 1 ($\frac{11}{3}$), Group I Row V 2b UV ($-\frac{11}{6}$) and Group II Row IV. Their analysis is unchanged (`Finite_terms_final.pdf`, Secs. 4, 5): the β_0 terms cancel, and Group II Row IV with the blue line is the DGLAP term.

2 Where the blue lines come from

Group III Rows III (both parts), V, VI and Group II Row IV have one extra (blue) line each. It is $-\log\frac{\vee}{k^+}$ times the BFKL term that is contained in line 1 of the row. Line 1 is kept unevaluated here, and the same term is in the BFKL list of `LL_terms.pdf`. Without the blue line it would be counted twice.

2.1 The short answer: do the p^+ integral first

In Group III Rows III, V, VI there are *two* different $\log\frac{\vee}{k^+}$:

1. the one you wrote, in line 2, $\left[\log\frac{k^+}{\Lambda} + \log(\frac{\vee}{k^+} - 1)\right]$. It is cancelled by the $f(0)$ subtraction of your $\left[\frac{1}{p^+}\right]_+$ in line 1 (Sec. 2.2). It is *not* a BFKL term;
2. one you did not write. It sits in line 1, in the term $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ of the bracket, and only shows up after the y' integral. This one *is* the BFKL term.

The $-\frac{1}{k^+}$ term has no $\frac{1}{p^+}$, so its p^+ integral needs no plus prescription and can be done first. With $r = y' - z$, $\lambda = \frac{\vee}{k^+}$:

$$\int_0^{\vee - k^+} dp^+ \, e^{-ik\cdot r \frac{p^+}{k^+}} = \frac{k^+}{i\,k\cdot r}\Big[1 - e^{-i(\lambda-1)\,k\cdot r}\Big], \qquad \text{and with } e^{-ik\cdot(y'-w)} = e^{-ik\cdot(z-w)}e^{-ik\cdot r}: \qquad \frac{k^+}{i\,k\cdot r}\Big[e^{-ik\cdot r} - e^{-i\lambda\,k\cdot r}\Big].$$

Multiplied by the kernel $\frac{r^m}{r^2}$ the y' integrand behaves like $\frac{1}{r^2}$ near $y' = z$. $\int \frac{d^2r}{r^2}$ diverges logarithmically, and the bracket cuts it off on both sides:

- for $r \ll \frac{k^+}{\vee|k|}$ the two exponentials cancel;
- for $\frac{k^+}{\vee|k|} \ll r \ll \frac{1}{|k|}$ the second one oscillates away and the first is ≈ 1: this range gives $\int \frac{dr}{r} = \log\frac{\vee}{k^+}$;
- for $r \gg \frac{1}{|k|}$ both oscillate away.

Exactly, after the angular integral, $\int_0^\infty \frac{dr}{r}\big[J_0(|k|r) - J_0(\lambda|k|r)\big] = \log\lambda = \log\frac{\vee}{k^+}$. With a smooth factor $G(r) = e^{-r^2/2}$ (`direct_check.py`): $\int_0^\infty \frac{dr}{r}\big[J_0(r) - J_0(\lambda r)\big]G(r) - \log\lambda = -0.27989,\ -0.27989,\ -0.27989,\ -0.27989$ for $\lambda = 10, 10^2, 10^3, 10^4$, i.e. it grows exactly like $\log\lambda$. Hence

$$-\frac{1}{k^+}\int_0^{\vee-k^+} dp^+ \int_{y'} e^{-ik\cdot(y'-w)}e^{-ik\cdot(y'-z)\frac{p^+}{k^+}}\frac{(y'-z)^m}{(y'-z)^2}G_m(y') = -\log\frac{\vee}{k^+}\int_{y'} e^{-ik\cdot(y'-w)}\frac{(y'-z)^m}{(y'-z)^2}G_m(z) + \text{finite},$$

which is the term (B). It is a *transverse* logarithm, from $y' \to z$ with $\frac{k^+}{\vee|k|} \lesssim |y'-z| \lesssim \frac{1}{|k|}$; pointwise in y' there is no logarithm. Since $|y'-z| \sim \frac{k^+}{p^+|k|}$, it is the region $k^+ \ll p^+ \ll \vee$: the unobserved gluon is harder than the measured one, the parent sits at $y' = \frac{k^+u + p^+z}{k^+ + p^+} \to z$, and the measured gluon is emitted by the gluon at z. This is the BFKL evolution of the projectile between the rapidity of the measured gluon and $\vee$. (The JIMWLK logarithm is the other end, $p^+ \ll k^+$, from your $f(0)$ subtraction.) The other terms of the bracket ($\frac{1}{p^+ + k^+}$, and $\frac{f(p^+)}{p^+}$ of $\left[\frac{1}{p^+}\right]_+$) carry an extra $\frac{1}{p^+}$, so their integrand is only $\frac{\log r}{r}$ near $y' = z$: no logarithm of $\vee$.

Group II Row IV, the same in your variable ξ: with $t = \frac{1}{\xi} \in [1, \frac{\vee}{k^+}]$ the term $\frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}$ is $(t-1)\,dt$ and the phase is $e^{-ik\cdot(y'-y)t}$. The $-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}$ of your bracket has $\int d^2\Delta\, e^{-ik\cdot\Delta t}\log\Delta^2 = -\frac{4\pi}{k^2t^2}$, so ($d_\perp = 2$)

$$\frac{4\pi}{k^2}\int_1^{\vee/k^+} dt\,\frac{t-1}{t^2} = \frac{4\pi}{k^2}\Big[\log\frac{\vee}{k^+} - 1 + \frac{k^+}{\vee}\Big],$$

from $y' \to y$ ($|y'-y| \sim \frac{1}{|k|t}$), with $y, y' \to z$: that is $\log\frac{\vee}{k^+}H_4$. The constant parts of the bracket (pole, γ_E, -1) multiply a smooth function of $y'-y$, whose Fourier transform falls fast in t: no logarithm.

2.2 Group III Row III, first part (Rows V and VI are identical in structure)

Line 1 and line 2. Write line 1 of your row as

$$L_1 = \frac{C}{2\pi}\int_0^{\vee-k^+} dp^+\Big[\frac{F_1(p^+)}{p^+ + k^+} - \frac{F_2(p^+) - F_2(0)}{p^+} - \frac{F_3(p^+)}{k^+}\Big], \qquad C = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}, \tag{2.1}$$

with your plus prescription $\left[\frac{1}{p^+}\right]_+ f(p^+) = \frac{1}{p^+}\big(f(p^+) - f(0)\big)$ and

$$F_n(p^+) = \int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\,T_n\,\mathcal{C}(y'), \tag{2.2}$$

$T_1 = \delta_{k'm}\delta_{ij}$, $T_2 = \delta_{jm}\delta_{ik'}$, $T_3 = \delta_{im}\delta_{jk'}$, and $\mathcal{C}(y')$ your colour factor (it contains $U^{e'c'}(y')$). $F_2(0)$ has no phase: it is the integrand of your line 2, which is the $\frac{F_2(0)}{p^+}$ part integrated from Λ,

$$L_2 = -\frac{C}{2\pi}\,F_2(0)\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+} = -\frac{C}{2\pi}\,F_2(0)\Big[\log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+} - 1\Big)\Big]. \tag{2.3}$$

The large-p^+ behaviour. The phase has the momentum $q = k\,\frac{p^+ + k^+}{k^+}$. With the Fourier transform $\int d^2r\,e^{-iq\cdot r}\frac{r^m}{r^2}G(r) = -2\pi i\frac{q^m}{q^2}G(0) + O(\frac{1}{q^2})$ (checked numerically in `BFKL_final.pdf`: ratio 1.00000, 1.00001, 1.00005, 1.0002 at $q = 6k, 11k, 21k, 41k$),

$$F_n(p^+) = \frac{k^+}{p^+ + k^+}\,F_n^z + O\Big(\frac{1}{p^{+2}}\Big), \qquad F_n^z = \int e^{-ik\cdot(y'-w)}\,\frac{(y'-z)^m}{(y'-z)^2}\Big[\frac{(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(x'-y')^2\,(y-w)^2\,(x-z)^2}\,T_n\,\mathcal{C}(y')\Big]_{y'\to z}, \tag{2.4}$$

i.e. $|y'-z| \lesssim \frac{k^+}{p^+|k|}$: for $p^+ \gg k^+$ the parent of the measured gluon sits on the soft gluon z, and the measured gluon is emitted by it. This is the projectile evolution between k^+ and $\vee$.

Term by term.

- $\frac{F_1}{p^+ + k^+} = O(\frac{1}{p^{+2}})$: no logarithm.
- $\left[\frac{1}{p^+}\right]_+$: the part $-\frac{F_2(p^+)}{p^+} = O(\frac{1}{p^{+2}})$ gives no logarithm. The subtraction $+\frac{F_2(0)}{p^+}$ does: exactly,

$$-\int_0^{\vee - k^+} dp^+ \frac{F_2(p^+) - F_2(0)}{p^+} = -\int_0^{k^+} dp^+ \frac{F_2(p^+) - F_2(0)}{p^+} - \int_{k^+}^{\vee - k^+} dp^+ \frac{F_2(p^+)}{p^+} + F_2(0) \log\Big(\frac{\vee}{k^+} - 1\Big).$$

The last term cancels the $\log(\frac{\vee}{k^+} - 1)$ of L_2, and the middle integral is finite for $\vee \to \infty$. So $L_1|_{T_2} + L_2 = -\frac{C}{2\pi} F_2(0) \log \frac{k^+}{\Lambda} +$ finite. **This is the hidden logarithm**: the log $\frac{\vee}{k^+}$ of line 2 is not a BFKL term. Your plus prescription removes the soft end $p^+ \to 0$ (Λ, JIMWLK) only.

- $-\frac{F_3(p^+)}{k^+}$ does not fall off: by Eq. (2.4) it is $-\frac{F_3^z}{p^+ + k^+}$ at large p^+. Since $\int_0^{\vee - k^+} \frac{dp^+}{p^+ + k^+} = \log \frac{\vee}{k^+}$ exactly,

$$-\int_0^{\vee - k^+} dp^+ \frac{F_3(p^+)}{k^+} = -\int_0^{\vee - k^+} \frac{dp^+}{k^+} \Big[F_3(p^+) - \frac{k^+}{p^+ + k^+} F_3^z\Big] - F_3^z \log \frac{\vee}{k^+},$$

and the first integral is finite for $\vee \to \infty$. **This is the BFKL term of the row**, hidden in line 1:

$$-\frac{C}{2\pi} F_3^z \log \frac{\vee}{k^+} = -\frac{1}{(2\pi)^3} \frac{ig^4 f^{c'd'a}}{16\pi^5} \frac{1}{k^+} \log \frac{\vee}{k^+} \int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)} \frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\, \mathcal{C}(z), \quad (2.5)$$

the term (B) of `LL_terms.pdf` (T_3 contracts $(y'-z)\cdot(y-w)$ and $(x'-y')\cdot(x-z) \to (x'-z)\cdot(x-z)$; $U^{e'c'}(y') \to U^{e'c'}(z)$).

So the row is

$$\frac{d^3N}{d^3k} = \underbrace{\log \frac{k^+}{\Lambda}\Big(-\frac{C}{2\pi} F_2(0)\Big)}_{\text{JIMWLK}} + \underbrace{\log \frac{\vee}{k^+}\Big(-\frac{C}{2\pi} F_3^z\Big)}_{\text{BFKL (B)}} + \text{finite}. \quad (2.6)$$

The finite terms. Here the row is written as L_1 (unevaluated) + line 2 with $\log(\frac{\vee}{k^+} - 1)$ only + the blue line $+\frac{C}{2\pi} F_3^z \log \frac{\vee}{k^+} = -$(B). By the two equations above this is finite for $\vee \to \infty$; the blue line is a subtraction at the hard end $p^+ \to \vee$, as $\left[\frac{1}{p^+}\right]_+$ is a subtraction at the soft end $p^+ \to 0$. Numerically (`finite_final/checks_xi.json`, a model F_3 with the tail of Eq. (2.4)), for $\vee = 10^2, 10^4, 10^6$: the $-\frac{1}{k^+}$ term alone gives -5.13, -9.75, -14.35 (it drops by $\log 100 = 4.61$ per step); with the blue line it gives -0.790, -0.800, -0.800.

The term (B) is also needed for BFKL: without it, 9 colour structures of the two-ρ $\log \frac{\vee}{k^+}$ sum differ from BFKL⊗LO, all with kernels in which the measured gluon is emitted by z, such as $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)$ (`BFKL_final.pdf`).

2.3 Group III Row III, second part

The same, with $\left[\frac{1}{p^+}\right]_+$ for the hidden logarithm. The term that does not fall off is

$$-\frac{p^+}{k^{+2}} \left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right] \frac{1}{p^+\,(x-z)^2 + k^+\,(x-w)^2} \xrightarrow{p^+ \gg k^+} -\frac{1}{k^{+2}\,(x-z)^2}\Big[\cdots\Big],$$

which with Eq. (2.4) gives $\log \frac{\vee}{k^+}$ times (B) of Row III part 2. All other terms of the bracket fall like $\frac{1}{p^+}$ and give no logarithm.

2.4 Group II Row IV

Your line 1 is, with $\xi = \frac{k^+}{p^+ + k^+}$ ($\xi \to 1$: $p^+ \to 0$; $\xi \to \frac{k^+}{\vee}$: $p^+ \to \vee$) and $E(y'-y)$ your pole bracket,

$$A \int_{x,y,x',y'} \frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\, E(y'-y) \int_{\frac{k^+}{\vee}}^1 \frac{d\xi}{\xi^2}\, e^{-ik\cdot(y'-y)\frac{1}{\xi}} \left[\xi(1-\xi) + \frac{(1-\xi)}{\xi} + \frac{\xi}{(1-\xi)_+}\right] \mathcal{C}, \qquad A = \frac{g^2}{4\pi^2} \frac{N_c}{(2\pi)^3} \frac{g^2}{2\pi^2} \frac{1}{k^+}.$$

- $\frac{d\xi}{\xi^2} \frac{\xi}{1-\xi} = \frac{d\xi}{\xi(1-\xi)} = -\frac{dp^+}{p^+}$: your $\frac{\xi}{(1-\xi)_+}$ is your $\left[\frac{1}{p^+}\right]_+$, with $f(0)$ the value at $\xi = 1$ (phase $e^{-ik\cdot(y'-y)}$). Its subtraction integrated from $\xi = \frac{k^+}{\Lambda + k^+}$ gives line 2, and integrated down to $\xi = \frac{k^+}{\vee}$ it gives $-\log(\frac{\vee}{k^+} - 1)$, which cancels that of line 2: the hidden logarithm, as above.
- The term $\frac{1-\xi}{\xi}$ does not fall off ($\frac{d\xi}{\xi^2} \frac{1-\xi}{\xi} = -\frac{p^+ dp^+}{k^{+2}}$). At small ξ the phase has the momentum $\frac{k}{\xi}$ and probes $|y'-y| \lesssim \frac{\xi}{|k|}$. With $\int d^2r\, e^{-iq\cdot r} \log r^2 = -\frac{4\pi}{q^2}$, the $-\frac{d_\perp}{2}$ $\log \frac{(y'-y)^2\mu^2}{4}$ of E gives $\frac{2\pi d_\perp \xi^2}{k^2}$, so the integrand is $\propto \frac{d\xi}{\xi}$ and gives $\log \frac{\vee}{k^+}$, with $y, y' \to z$. With $\frac{4\pi}{k^2} = \frac{1}{\pi} \int_{w,w'} e^{-ik\cdot(w'-w)} \mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$ and $\frac{A}{\pi} = \frac{1}{(2\pi)^3} \frac{g^4 N_c}{8\pi^5} \frac{1}{k^+}$ this is

$$\log \frac{\vee}{k^+}\, H_4, \qquad H_4 = \frac{1}{(2\pi)^3} \frac{g^4 N_c}{8\pi^5} \frac{1}{k^+} \int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)} \frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2} \frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2} \times \big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ce}(x)\big] \rho^{e'}(x') \rho^e(x) :$$

both the measured gluon and its conjugate are emitted by the gluon at z. It is the $-2\,\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)$ line of BFKL⊗LO. The terms $\xi(1-\xi)$ and $\frac{\xi}{(1-\xi)_+} h(\xi)$ give $\propto \xi\, d\xi$: no logarithm.

- The blue line $-\log\frac{\vee}{k^+}H_4$ takes it out of the finite terms, since it is in the BFKL list (C1). Lines 1, 2 and the blue line are then the DGLAP term (`Finite_terms_final.pdf`, Sec. 5).

3 Group I (with c.c.)

3.1 Row I

Only four ρ's (`LL_terms.pdf`, Sec. 6); no finite terms in your files.

3.2 Row II

Row II, first part. The pole bracket times $\frac{11}{3}$ is the β_0 term; it cancels against the $-\frac{11}{6}$ of Row V 2b UV (`Finite_terms_final.pdf`, Sec. 4). $\log(1-\frac{k^+}{\vee})\to 0$ for $\vee\to\infty$.

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x'}\!\!e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times\right.\\ &\left[\frac{11}{3}+\log\left(1-\frac{k^+}{\vee}\right)\right]-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\Bigg]\times\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\Big]\end{aligned}\tag{3.1}$$

Row II, second part, d^3N_{2a}. Nothing is left.

Row II, second part, d^3N_{2b}. Unchanged: no large logarithm. R^i as in your file.

$$\begin{aligned}\frac{d^3N_{2\,b}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'}\!\!e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y}\!\!e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})\\ &\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\left\{\rho^b(x),\rho^c(y)\right\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\left\{\rho^d(x),\rho^e(y)\right\}\Big]\end{aligned}\tag{3.2}$$

3.3 Row III

Row III, d^3N_1. Red: the $\frac{1}{\Lambda}$ lines (your green brackets). Lines 1 and 4 here, line 1 of d^3N_2 and line 1 of d^3N_{3a} add up to zero exactly, pointwise (`check_split.py`: 3.0×10^{-16}).

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{w,w'}\!\!e^{-ik(w'-w)}\Bigg\{\\ &\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}+\frac{1}{k^+-\vee}\right]\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &-\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right]\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &+\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{\frac{k^+}{\vee}\,(x-w)^2-(x-z)^2}{(x-w)^2}\right]\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}\end{aligned}\tag{3.3}$$

Row III, d^3N_2. Red: line 1 is a $\frac{1}{\Lambda}$ line (see d^3N_1); your two red lines cancel each other.

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{\\
&\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,[(x-w)^2-(x-z)^2]}\left[\frac{1}{(\vee-k^+)}-\frac{1}{\Lambda}\right]\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\Bigg\}
\end{aligned}
\tag{3.4}
$$

Row III, d^3N_{3a}. Red: line 1 is a $\frac{1}{\Lambda}$ line (see d^3N_1). The edge-kernel line has no finite part: its $\log\frac{k^+-\Lambda}{\Lambda} = \log\frac{k^+}{\Lambda}+\log(1-\frac{\Lambda}{k^+})$ and $\log(1-\frac{\Lambda}{k^+})\to 0$. $\mathcal{A}_0$ is $\mathcal{A}$ at $\Lambda=0$, Eq. (1.1).

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{\\
&\frac{ig^4\,f^{abd}}{8\pi^5}\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2}
\end{aligned}
$$

$$
\begin{aligned}
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}_0\int_{x,y,z,x'} \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}_0\int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}_0\int_{x,y,z,x'} \frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
\end{aligned}
\Bigg\} \tag{3.5}
$$

Row III, d^3N_{3b}. **C4.** The last line (red) is, for $\Lambda \to 0$, $\log\frac{(z-w)^2}{(x-w)^2}$ times the edge kernel $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$: not integrable at $z \to w$. The edge kernel is replaced by $\mathcal{D}$ (LL file), and its finite part belongs to the region of $\mathcal{D}$, which is not yet computed (Q3, open).

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}_0\int_{x,y,z,x'} \frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(z-w)^2}{(x-w)^2}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(z-w)^2}{(x-w)^2}\right]\int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(z-w)^2}{(x-w)^2}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
\end{aligned}
\Bigg\} \tag{3.6}
$$

3.4 Row IV

Row IV.

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\Bigg\{ \\
&\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right] \times \left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
&\left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right]
\end{aligned}
$$

$$+\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(1-\frac{k^+}{\vee}\right)\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big] \tag{3.7}$$

3.5 Row V

Row V, first part.

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\left[\delta_{im}\,\delta_{jk}-\left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times$$
$$\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]$$
$$-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\left[\frac{1}{q^+}\right]_+$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big] \tag{3.8}$$

Row V, second part, d^3N_{2a}.

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\times$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{3.9}$$

Row V, second part, $d^3N_{2b\,\mathrm{UV}}$. Only $-\frac{11}{6}$ is left (E1, $2\log(1-\frac{\Lambda}{k^+})\to 0$). It is the β_0 term that cancels the $\frac{11}{3}$ of Row II part 1.

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[-\frac{11}{6}\right]\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y). \tag{3.10}$$

Row V, second part, $d^3N_{2b\,\mathrm{finite}}$. $\Lambda\to 0$ in the limits of the plus-prescribed integrals (regular at the end points). In line 6, $2\log\frac{k^+-\Lambda}{\Lambda}$ went to the LL file; $-\frac{11}{6}$ is the counterterm constant.

$$\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}$$
$$\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+(k^+-p^+)}{k^{+2}}\right.$$
$$\left.+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\right.$$

$$
\begin{aligned}
&- \left[\frac{k^+}{p^+}\right]_+ \left[\frac{(y-z)^i (x-z)^m}{(x-z)^2} + \frac{(y-x)^m (y-z)^i}{2(y-x)^2}\right] + \left[\frac{(y-z)^i (x-z)^m}{(x-z)^2} + \frac{(y-x)^m (y-z)^i}{2(y-x)^2}\right] \\
&+ \frac{p^+ \delta_{im}}{k^+} \left[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2(y-z)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right] \\
&- \left[\frac{(y-x)^m (x-z)^i}{(x-z)^2} - \frac{(y-x)^m (y-z)^i}{2(y-z)^2} + \frac{(y-z)^m (x-z)^i}{(x-z)^2} + \frac{(y-x)^i (y-z)^m}{2(y-x)^2}\right]\Bigg] \\
&\times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\, U^{af}(y) - N_c\, U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y) \\
&+ \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0 \frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+(y-z)^2]} \\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+ \times \\
&\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\, U^{af}(y) - N_c\, U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y) \\
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0 \frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+(y-z)^2]} \\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right] \\
&\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\, U^{af}(y) - N_c\, U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y) \\
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}\right] \\
&\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2} \times \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\, U^{af}(y) - N_c\, U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{3.11}
$$

4 Group II

4.1 Row I

Row I, first part, first line.

$$
\begin{aligned}
\frac{d^3 N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{ \\
&- \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times \\
&\left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right] \\
&- \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\} \\
&\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\} - U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{4.1}
$$

Row I, first part, second line.

$$
\begin{aligned}
\frac{d^3 N}{d^3k} &= \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{ \\
&\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}(x-w')^2}{(x-w')^2}\right]\times \\
&\left[\frac{\delta^{ij}}{2\,(z-w')^2} + \frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2} - \frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2} - \frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right] \\
&+ \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2} + \frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\} \\
&\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x) - f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\} - U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]
\end{aligned}
\tag{4.2}
$$

Row I, second part, d^3N_{2a}.

$$
\frac{d^3 N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\Bigg\{
$$

$$
\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times
$$
$$
\left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right.
$$
$$
+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}
$$
$$
+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}
$$
$$
\left.-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\right]\times
$$
$$
\left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{(x-w)^2}{(x'-w')^2}\right]\right]\times
$$
$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$
$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{4.3}
$$

Row I, second part, d^3N_{2b}.

$$
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2\right]}\times
$$
$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$
$$
\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times
$$
$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$
$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$
$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{\left[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2\right]}\times
$$
$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$
$$
\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times
$$
$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$
$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$
$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{\left[(x'-w')^4\,(x-z)^2\,-(x-w)^2\,(x'-z)^2\,(x'-w')^2\right]}\times
$$
$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$
$$
\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]
$$
$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$
$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{4.4}
$$

Row I, second part, d^3N_{2c}.

$$
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}
$$

$$
\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{4.5}
$$

Row I, second part, d^3N_{2d}.

$$
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k}=+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{4.6}
$$

Row I, second part, d^3N_{2e}.

$$
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times
$$

$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\Big[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}
$$

$$
+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Big]\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{(x'-w')^2}\right]\times
$$

$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{4.7}
$$

Row I, second part, d^3N_{2f}. $\frac{1}{k^+ + \Lambda} \to \frac{1}{k^+}$.

$$
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,[(x-z)^2-(x-w)^2]}\,\frac{\left[(x'-z)^2-(x'-w')^2\right]}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times
$$

$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\left[\frac{1}{\vee}-\frac{1}{k^+}\right]\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}
$$

$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}
$$

$$
\frac{[(x-z)^2-(x-w)^2]}{[(x'-z)^2-(x'-w')^2]\,[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{(x'-w')^2}\right]\times
$$

$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{4.8}
$$

4.2 Row II

Only four ρ's (`LL_terms.pdf`, Sec. 6); no finite terms in your files.

4.3 Row III

Only four ρ's (`LL_terms.pdf`, Sec. 6); no finite terms in your files.

4.4 Row IV

Row IV. C1. The blue line, $-\log\frac{\vee}{k^+}H_4$, is the BFKL term contained in the ξ integral of line 1; it is now in the BFKL list. Line 2 keeps $\log(\frac{\vee}{k^+}-1)$, which cancels the $\log\frac{\vee}{k^+}$ of the $\frac{\xi}{(1-\xi)_+}$ integral. Lines 1, 2 and the blue line are the DGLAP term

(Finite_terms_final.pdf, Sec. 5). In line 3, $\frac{k^+}{k^+ + \Lambda} \to 1$.

$$
\begin{aligned}
\frac{d^3 N_{\rm NLO}}{d^3 k} &= \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} \frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[-\frac{d_\perp}{\epsilon} - d_\perp\,\gamma_E - \frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right) - 1\right] \\
&\left[\int_{\frac{k^+}{\vee}}^1 \frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\left[\xi\,(1-\xi) + \frac{(1-\xi)}{\xi} + \frac{\xi}{(1-\xi)_+}\right]\right] \times \left[U^{e'c}(y') - U^{e'c}(x')\right] \times \left[U^{ec}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \\
&+ \frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left(\frac{\vee}{k^+} - 1\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[-\frac{d_\perp}{\epsilon} - d_\perp\,\gamma_E\right. \\
&\left. - \frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right) - 1\right]\,\left[U^{e'c}(y') - U^{e'c}(x')\right] \times \left[U^{ec}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \\
&+ \frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2} - \frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\times \\
&\left[U^{e'c}(y') - U^{e'c}(x')\right] \times \left[U^{ec}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \\
&- \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2} \\
&\left[U^{e'c}(z) - U^{e'c}(x')\right] \times \left[U^{ec}(z) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)
\end{aligned}
\tag{4.9}
$$

5 Group III (with c.c.)

5.1 Row I

Row I.

$$
\begin{aligned}
\frac{d^3 N_{\rm NLO}}{d^3 k} &= \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg\{ \\
&- \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right] \times \left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
&\left. - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right] \\
&- \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\
&\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times \\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{5.1}
$$

5.2 Row II

Row II.

$$
\begin{aligned}
\frac{d^3 N_{\rm NLO}}{d^3 k} &= \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg\{ \\
&- \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right] \times \left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
&\left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right] \\
&- \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\
&\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right] \times \left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{5.2}
$$

5.3 Row III

Row III, first part. C3. The blue line is $-\log\frac{\vee}{k^+}\times$(B): the BFKL term of the $\left[\frac{1}{p^+}\right]_+$ integral (region $y' \to z$), now in the BFKL list. Line 2 keeps $\log(\frac{\vee}{k^+} - 1)$, which cancels the $-\log\frac{\vee}{k^+}$ of the subtraction in $\left[\frac{1}{p^+}\right]_+$. Lines 1, 2 and the blue line

together are finite for $\vee \to \infty$.

$$
\begin{aligned}
\frac{d^3 N_{1\,\mathrm{NLO}}}{d^3 k} &= \frac{1}{(2\pi)^3}\,\frac{i g^4\, f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee - k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\, e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+ \delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]\\
&- \frac{1}{(2\pi)^3}\,\frac{i g^4\, f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}-1\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]\\
&+ \frac{1}{(2\pi)^3}\,\frac{i g^4\, f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\
&\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\\
&\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]
\end{aligned}
\tag{5.3}
$$

Row III, second part. C3 as in the first part. **C5**: measure.

$$
\begin{aligned}
\frac{d^3 N_{2\,\mathrm{NLO}}}{d^3 k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee - k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\, e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2 + k^+\,(x-w)^2]}\\
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^+ + k^+)^2}\,[(x-z)^2-(x-w)^2] + \frac{p^+\,\delta_{k'm}}{k^+\,(p^+ + k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2} - \frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.\\
&\left.\left. - \frac{[(x-z)^2-(x-w)^2]}{2\,(z-w)^2}\right] + \frac{\delta_{k'm}}{(p^+ + k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2\,(x-z)^2} - \frac{[(x-z)^2-(x-w)^2]}{2\,(z-w)^2}\right]\right.\\
&- \left[\frac{1}{p^+}\right]_+\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2} + \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right] - \frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\\
&\left.- \frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2} + \frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2} + \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}-1\right)\int_{w,y',z,y,x}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.\\
&\left.+ \frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&+ \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}\\
&\times\left[\frac{(y'-z)\cdot(z-w)}{(y'-z)^2\,(z-w)^2} - \frac{(y'-z)\cdot(x-w)}{2\,(y'-z)^2\,(x-w)^2}\right]\\
&\left[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\\
&\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{5.4}
$$

5.4 Row IV

Only four ρ's (`LL_terms.pdf`, Sec. 6); no finite terms in your files.

5.5 Row V

Row V. C3 as in Row III.

$$
\begin{aligned}
\frac{d^3 N_{\mathrm{NLO}}}{d^3 k} &= \frac{1}{(2\pi)^3}\,\frac{i g^4\, f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee - k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\, e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+ \delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]\\
&- \frac{1}{(2\pi)^3}\,\frac{i g^4\, f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}-1\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]
\end{aligned}
$$

$$
+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}
$$
$$
\left[U^{e'c'}(z)\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\left[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\right] \tag{5.5}
$$

5.6 Row VI

Row VI. C3 as in Row III.

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$
$$
\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times
$$
$$
\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x)-U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]
$$
$$
-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}-1\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$
$$
\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x)-U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]
$$
$$
+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}
$$
$$
\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]
$$
$$
\times\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x)-U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right] \tag{5.6}
$$

6 Checks

`check_split.py`: for each of the 79 terms without a remaining p^+, ξ or momentum integral, your term = (its log $\frac{k^+}{\Lambda}$ part) + (its log $\frac{\vee}{k^+}$ part) + (the exact remainder) to 1.5×10^{-14}, at 3 configurations and $(k^+,\vee,\Lambda)=(1,37,0.013),(0.6,900,3\times10^{-4})$; the remainder written here is its $\Lambda\to0$ limit (3.5×10^{-8} at $\Lambda=10^{-9}$). The terms with a remaining integral (Group I Row II 2b, Row V; Group II Row IV lines 1, 3; Group III line 1 of Rows III, V, VI) have no Λ or large logarithm outside the limits treated above, and are copied.